Assessment 2 Portfolio

A proposed UK venture commercialising cold-atom gravity gradiometry for subsurface survey.

Market Opportunity Navigator

Part 1 – Market Opportunity Set

Assess the unique abilities to identify market opportunities for your business.

Core Abilities

List your business' core abilities or technological elements. Characterise them based on their functions and properties, independent of your product or service.

Core Ability 1

Description: Measures the gravitational difference between two clouds of laser-cooled atoms falling in one apparatus. It reads density contrast, so it needs no line of sight, no contact and no excavation.

Core Ability 2

Description: Vibration is common to both clouds and cancels differentially, shielding handles the rest. This took the instrument off the optical bench and onto a street (Stray et al., 2022).

Core Ability 3

Description: Bayesian inversion turns sparse gradient readings into a three-dimensional density estimate, reporting confidence in position and depth rather than one certain line.

Core Ability 4

Description: Packaging lasers, vacuum and shielding into equipment a trained surveyor can run through a working day. Most of it is tacit knowledge, which matters for value capture.

Market Opportunities

Identify the market opportunities. What applications can the core abilities be used in? Which customers or market segments may need them?

Opportunity 1

Application: Targeted pre-excavation survey of high-risk urban sites, finding buried assets, voids and abandoned structures that electromagnetic location and radar miss.

Customer Group: UK water, gas, power and telecoms asset owners, the tier-1 contractors digging for them, and highway and rail authorities.

Opportunity 2

Application: Density targeting for mineral exploration and geothermal assessment, narrowing where to drill before anyone drills.

Customer Group: Exploration companies, geothermal developers, and the geophysical survey contractors who serve them.

Opportunity 3

Application: Drift-free inertial navigation where satellite signals are jammed, spoofed or unavailable.

Customer Group: The Ministry of Defence and defence primes first, maritime and aviation operators later.

Part 2 – Market Attractiveness

Evaluate and compare options systematically.

The scorecard below is completed once for each of the three opportunities identified in Part 1.

Opportunity Name: Opportunity 1 — Pre-excavation subsurface survey

Impact

Problem Severity:

- Extent of concern
- Significance of consequences
- Number of people affected

Low ☐

Medium ☐

High ☒

Very High ☐

Pertinent Solution:

- Effectiveness and efficiency
- Better than alternatives
- Avoids secondary harm

Low ☐

Medium ☐

High ☒

Very High ☐

Impact Reach:

- Ease of measuring and reporting
- Enduring effect

- Scalability

Low ☐

Medium ☐

High ☒

Very High ☐

Overall Impact:

Low ☐

Medium ☐

High ☒

Very High ☐

Notes: One hole in every 65 hits something: about 60,000 strikes a year and £2.4bn in economic cost (Utility Strike Avoidance Group, 2023). Workers are hurt.

Potential

Compelling Reasons to Buy:

- Unmet need
- Effective solution
- Better than current solutions

Low ☐

Medium ☐

High ☒

Very High ☐

Potential Market Volume:

- Current market size
- Expected growth

Low ☐

Medium ☒

High ☐

Very High ☐

Economic Viability:

- Margins (Value vs Cost)
- Customer ability to pay
- Customer stickiness

Low ☐

Medium ☐

High ☒

Very High ☐

Overall Potential

Low ☐

Medium ☐

High ☒

Very High ☐

Notes: Contractors already buy PAS 128 surveys, so a budget line exists to sell against (British Standards Institution, 2022), and reported strike rates mean improvement can be shown, not claimed.

Challenge

Implementation Obstacles:

- Product development difficulties
- Funding challenges
- Sales and distribution difficulties

Low ☐

Medium ☒

High ☐

Very High ☐

Time to Revenue:

- Development time
- Time between product and market readiness
- Length of sale cycle

Low ☐

Medium ☒

High ☐

Very High ☐

External Risks:

- Competitive threat
- Barriers to adoption
- Third party dependencies

Low ☐

Medium ☐

High ☒

Very High ☐

Overall Challenge

Low ☐

Medium ☒

High ☐

Very High ☐

Notes: The physics already works outdoors, so this is productisation. Two risks bite: PAS 128 recognises radar and electromagnetic location, not gravity, and Delta.g, a Birmingham spin-out with £4.6m of seed funding, is ahead of us (University of Birmingham, 2025).

Opportunity Category

Gold Mine (High potential, low challenge) ☒

Moon Shot (High potential, high challenge) ☐

Quick Win (Low challenge, low potential) ☐

Questionnable (Low potential, high challenge) ☐

Notes: Gold Mine: highest potential, lowest challenge of the three, judged against each other, not in the abstract.

Opportunity Name: Opportunity 2 — Mineral and geothermal exploration

Impact

Problem Severity:

- Extent of concern
- Significance of consequences
- Number of people affected

Low ☐

Medium ☒

High ☐

Very High ☐

Pertinent Solution:

- Effectiveness and efficiency
- Better than alternatives
- Avoids secondary harm

Low ☐

Medium ☒

High ☐

Very High ☐

Impact Reach:

- Ease of measuring and reporting
- Enduring effect
- Scalability

Low ☐

Medium ☒

High ☐

Very High ☐

Overall Impact:

Low ☐

Medium ☒

High ☐

Very High ☐

Notes: Better targeting means fewer speculative boreholes, though the harm avoided is diffuse next to a struck gas main.

Potential

Compelling Reasons to Buy:

- Unmet need
- Effective solution
- Better than current solutions

Low ☐

Medium ☐

High ☒

Very High ☐

Potential Market Volume:

- Current market size
- Expected growth

Low ☐

Medium ☐

High ☒

Very High ☐

Economic Viability:

- Margins (Value vs Cost)
- Customer ability to pay
- Customer stickiness

Low ☐

Medium ☒

High ☐

Very High ☐

Overall Potential

Low ☐

Medium ☐

High ☒

Very High ☐

Notes: Global exploration budgets dwarf UK survey spend, and drilling costs enough that better targets are worth paying for.

Challenge

Implementation Obstacles:

- Product development difficulties
- Funding challenges
- Sales and distribution difficulties

Low ☐

Medium ☐

High ☒

Very High ☐

Time to Revenue:

- Development time
- Time between product and market readiness
- Length of sale cycle

Low ☐

Medium ☐

High ☒

Very High ☐

External Risks:

- Competitive threat
- Barriers to adoption
- Third party dependencies

Low ☐

Medium ☒

High ☐

Very High ☐

Overall Challenge

Low ☐

Medium ☐

High ☒

Very High ☐

Notes: Remote sites, power and calibration in harsh conditions sit beyond our packaging. Campaigns are seasonal, procurement slow, and airborne gradiometry already serves it.

Opportunity Category

Gold Mine (High potential, low challenge) ☐

Moon Shot (High potential, high challenge) ☒

Quick Win (Low challenge, low potential) ☐

Questionnable (Low potential, high challenge) ☐

Opportunity Name: Opportunity 3 — GPS-denied navigation

Impact

Problem Severity:

- Extent of concern
- Significance of consequences
- Number of people affected

Low ☐

Medium ☐

High ☒

Very High ☐

Pertinent Solution:

- Effectiveness and efficiency
- Better than alternatives
- Avoids secondary harm

Low ☐

Medium ☒

High ☐

Very High ☐

Impact Reach:

- Ease of measuring and reporting
- Enduring effect
- Scalability

Low ☐

Medium ☒

High ☐

Very High ☐

Overall Impact:

Low ☐

Medium ☒

High ☐

Very High ☐

Notes: Satellite navigation is jammed and spoofed routinely, with consequences from delayed shipping to lost life, the national mission targets quantum navigation on aircraft by 2030 (HM Government, 2023).

Potential

Compelling Reasons to Buy:

- Unmet need
- Effective solution
- Better than current solutions

Low ☐

Medium ☐

High ☒

Very High ☐

Potential Market Volume:

- Current market size
- Expected growth

Low ☐

Medium ☒

High ☐

Very High ☐

Economic Viability:

- Margins (Value vs Cost)
- Customer ability to pay
- Customer stickiness

Low ☐

Medium ☒

High ☐

Very High ☐

Overall Potential

Low ☐

Medium ☐

High ☒

Very High ☐

Notes: The state agrees the problem is severe, defence buyers pay well but are few and slow.

Challenge

Implementation Obstacles:

- Product development difficulties
- Funding challenges
- Sales and distribution difficulties

Low ☐

Medium ☐

High ☐

Very High ☒

Time to Revenue:

- Development time
- Time between product and market readiness
- Length of sale cycle

Low ☐

Medium ☐

High ☐

Very High ☒

External Risks:

- Competitive threat
- Barriers to adoption
- Third party dependencies

Low ☐

Medium ☐

High ☒

Very High ☐

Overall Challenge

Low ☐

Medium ☐

High ☐

Very High ☒

Notes: A survey instrument may stand still, a navigator must work while moving, at a fraction of the size, weight and power. Export control applies, and the dual-use question becomes ours to answer.

Opportunity Category

Gold Mine (High potential, low challenge) ☐

Moon Shot (High potential, high challenge) ☒

Quick Win (Low challenge, low potential) ☐

Questionnable (Low potential, high challenge) ☐

Part 3 – Agile Focus Strategy

Prioritise opportunities for a clear strategic focus.

Primary Opportunity: Opportunity 1, pre-excavation subsurface survey.

Other Opportunities: Opportunity 2, mineral and geothermal exploration, Opportunity 3, GPS-denied navigation.

Opportunity 1 — Agile Strategy: Pursue Now

Opportunity 2 — Mineral and geothermal exploration

Opportunity Assessment

Product Relatedness

To what extent do the products share technological competencies, required resources and necessary networks?

Low ☐

Medium ☐

High ☒

Market Relatedness

To what extent do the customers and markets share values and benefits, sales channels and possible risks?

Low ☒

Medium ☐

High ☐

Back-Up Option?

These should be attractive market opportunities that do not share major risks with the primary opportunity, and allow for a change in direction if needed.

Yes ☒

No ☐

Growth Option?

These should be attractive market opportunities that allow your business to create additional value in the future.

Yes ☐

No ☒

Agile Strategy:

Pursue Now ☐

Keep Open ☒

Place in Storage ☐

Opportunity 3 — GPS-denied navigation

Opportunity Assessment

Product Relatedness

To what extent do the products share technological competencies, required resources and necessary networks?

Low ☐

Medium ☒

High ☐

Market Relatedness

To what extent do the customers and markets share values and benefits, sales channels and possible risks?

Low ☒

Medium ☐

High ☐

Back-Up Option?

These should be attractive market opportunities that do not share major risks with the primary opportunity, and allow for a change in direction if needed.

Yes ☐

No ☒

Growth Option?

These should be attractive market opportunities that allow your business to create additional value in the future.

Yes ☒

No ☐

Agile Strategy:

Pursue Now ☐

Keep Open ☒

Place in Storage ☐

Notes: The kernel is Rumelt's (2011): our diagnosis is a laboratory-grade sensor with no route to a customer, so the guiding policy is to earn revenue on the ground everyone else struggles with, maturing the instrument on a client's site rather than our own. Exploration is a genuine back-up, its buyers unrelated to UK construction, so a stall in contractor adoption would not stop it. It is not a hedge against technical failure: all three need the same sensor to work, and pretending otherwise is the fiction this tool exists to prevent (Gruber and Tal, 2017). Keeping both open costs design headroom and a data-sharing agreement, but no engineers.

Value Proposition Canvas

Value Proposition

Notes: We tell contractors what lies under the ground where radar gives up. Stated honestly: slower per metre and coarser than radar in good conditions, so the fit is second pass on hard ground, not replacement of the first (Osterwalder et al., 2014). Claiming more would put crews at risk, which is where responsible innovation becomes a design constraint (Stilgoe, Owen and Macnaghten, 2013).

Products and Features

- What can help a customer get a job done?
- Does it address specific jobs or a variety?
- Are the features tangible or digital/virtual?

Notes: A survey service, not an instrument sale: a trolley or vehicle-mounted quantum gravity gradiometer run by our own crews. One specific job, the risk sign-off on difficult ground, not a variety. Features tangible and digital: the field service, and a depth-resolved density map with stated uncertainty, issued in GIS and BIM formats and cross-referenced to PAS 128 quality levels.

Gain Creators

- How do your features create savings in time, money and effort?
- Do they produce expected outcomes or exceed expectations?
- How do they outperform existing solutions?

Notes: Savings: priced by the day against the true cost of one strike, 29 times its repair bill (Utility Strike Avoidance Group, 2023) — arithmetic a buyer can run themselves. Expectations: on the ground it is built for it exceeds them, where current methods return a caveat, in good conditions it only meets them, and we say so. Outperformance: outcomes are reported against the national baseline, so the claim is checkable, and data is returned NUAR-ready, turning a survey cost into an asset the owner keeps.

Pain Relievers

- How do your features prevent frustrations and annoyances?
- How do they improve upon under-performing solutions?
- Do they mitigate risks or eliminate obstacles?

Notes: Frustrations: the sensor reads density contrast, so wet clay and plastic pipes stop being the problem they are for radar and electromagnetic location. Improvement: voids, culverts and abandoned shafts become visible, and each survey should remove several trial holes. Risk: the output carries explicit uncertainty (Stray et al., 2022), so the manager prices residual risk rather than discovering it with a digger. It mitigates rather than eliminates: PAS 128 mandates radar and electromagnetic location, admitting gravity only as a supplementary method.

Customer Segment

Notes: The survey or HSE manager at a UK utility asset owner, or the tier-1 contractor digging for them, who signs off pre-excavation risk. They are the buyer, the site crew is the user, the asset owner's insurer is a third party whose interests we can borrow.

Jobs

- How does your customer interact with your business?
- Do their activities and goals change in different contexts?
- What products or services are they currently using to complete these activities?

Notes: Interaction: commissioned per site as a second pass beside a PAS 128 contract, they receive a map, not an instrument. Context: always to know what is under the ground before breaking it, on a verge a compliance step, beside a gas main a safety sign-off that must be defensible. Current tools: record drawings, electromagnetic location, radar and trial holes, returned into NUAR.

Gains

- How do customers measure performance, quality and success?
- Would savings in time, money or effort be of the most value?
- What positive social consequences do they desire?

Notes: Measured by strike rate per thousand excavations, PAS 128 quality level, and days lost to stoppages. Time is the saving valued most: an emergency reinstatement closes a street for longer than the survey took. Socially: fewer injured workers, fewer disrupted streets, and data that stays useful through NUAR. They would pay most for a strike-rate improvement they can show an insurer and a board.

Pains

- What are the barriers to adoption of current solutions to the problem?
- What are the main challenges they face when interacting with current solutions?
- How are current value propositions underperforming?

Notes: Barriers: some 4 million km of buried pipes and cables, a hole dug every seven seconds (Geospatial Commission, 2024), and records that are often wrong, trial holes sample points only, at a cost in money and closures. Challenges: radar loses signal in wet, clay-rich ground and at depth, electromagnetic location cannot see plastic. Underperformance: surveys come back qualified B3 or B4 (British Standards Institution, 2022), handing the risk back to the manager. The worst outcome is not the invoice but the injury.

Business Model Canvas

Key Partners

- What suppliers do you need?
- What resources do they provide?
- What activities do they fulfil?
- Who else do you need to work with?
- Who are the other key stakeholders?

Notes: Suppliers: a university physics group for the licence, laser, vacuum and shielding vendors. Resources: the interferometry IP and absorptive capacity, components whose lead times become ours. Activities: training the people we hire, calibration and repair. Who else: a PAS 128 survey firm as channel, the Geospatial Commission through NUAR. Stakeholders: Innovate UK, the National Quantum Technologies Programme, HSE and unions, engaged while the product is still shapeable: the triple helix in practice (Etzkowitz and Leydesdorff, 2000).

Key Activities

- What does your value proposition require?
- What do your distribution channels require?
- What do your customer relationships require?

Notes: Value proposition: paid field trials with lead users on their worst sites rather than our best (von Hippel, 1986), inversion software, standards work with BSI so gravity becomes specifiable under PAS 128. Channels: training the partner’s surveyors, settling the dual-use question early, since the instrument that finds a culvert finds a tunnel: export-control classification comes first. Relationships: recruiting and keeping the ten or so people who can do this.

Value Proposition

- What problem are you solving?
- What do your customers need?
- What do only they want?

Notes: Problem: excavation on ground where radar and electromagnetic location fail. Need: depth-resolved subsurface maps with stated uncertainty, defensible after the event. What only they want: risk reduction per site that an insurer will recognise, so it is sold as that, not as quantum technology: nobody buys physics.

Relationships

- What kind of relationship do they expect?
- Are they one-off or repeat customers?
- How will you maintain the relationship?

Notes: Expected: technical and consultative, not transactional, the manager must understand the uncertainty figure before signing. Repeat rather than one-off, because difficult sites recur. Maintained by a named crew per account, a data licence that renews, and a joint review after each survey, doubling as our post-project review: the learning routine Tidd and Bessant (2021) treat as a dynamic capability.

Customer Segments

- Who are you creating value for?
- Who are your most important customers?
- Who else benefits from your business?

Notes: Value is created for UK utility asset owners and tier-1 contractors on high-risk urban sites, and secondarily for highway, rail and local authorities. Most important: the first asset owners to allow trials on live sites, because their strike data becomes our evidence. Who else benefits: the site crew, the insurer, and the public whose street stays open, who never appear on the invoice.

Key Resources

- What does your value proposition require?
- What do your distribution channels require?
- What do your customer relationships require?

Notes: Value proposition: the licensed IP matters less than the tacit build-and-align knowledge staff hold, the part a competitor cannot buy, and the dataset of subsurface signatures that improves the inversion with every job. Channels: the partner's PAS 128 accreditation and field logistics. Relationships: the crews, and access control designed in from the first survey, because a map of what lies beneath critical infrastructure is a liability worth stealing.

Channels

- How will you reach them?
- Which channels do they use?
- How do they all fit together?

Notes: Reach: direct technical selling into a handful of asset owners for the first paid trials, then framework agreements. Channels they use: accredited PAS 128 survey firms under existing frameworks, so we partner with one already holding the accreditation, relationships and logistics we lack. Fit: direct sales prove the method, the partner scales it. The channel is the complementary asset (Teece, 1986), building one from nothing would cost more than the sensor did.

Cost Structure

- What costs are essential?
- Which resources are most expensive?
- Which activities are most expensive?

Notes: Essential: people, instrument, fieldwork, compliance — a value-driven structure, not a cost-driven one (Osterwalder and Pigneur, 2010). Most expensive resources: physicists and field engineers, the largest, least compressible and scarcest cost, then the instrument's lasers, vacuum and shielding, and its vehicle. Most expensive activities: field trials and traffic management. Standards, certification and export-control compliance are small lines, easy to underestimate and awkward to retrofit.

Revenue Streams

- What are your customers willing to pay?
- How would they prefer to pay?
- How much revenue will each stream contribute?

Notes: Willing to pay: a day rate set against the cost of one strike, as this industry already buys. Preferred: per-site day rates under a framework, with a renewing licence for the processed dataset. Contribution, as a planning assumption rather than a forecast: surveys about two-thirds of early revenue, data licences and re-surveys a fifth, Innovate UK and quantum-mission grants the rest, costing no equity. Leasing only later, once know-how is no longer the product.

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